

DINESH KUMAR M

Software Engineer | Full Stack Engineer (Backend Focused)

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PROFESSIONAL SUMMARY

Software Engineer with **3+ years of software development experience** building and supporting production applications serving **20K+ registered users**. Experienced in **Java, Spring Boot, Rust, PostgreSQL, DynamoDB, Redis, REST, GraphQL, gRPC, AWS, microservices, and distributed systems**, with production frontend experience in **Next.js and Angular**. Contributed to feature-level architecture, API and database design, asynchronous workflows, frontend integration, deployment validation, and production troubleshooting across **25+ releases**.

TECHNICAL SKILLS

Languages: Java, Rust, TypeScript, JavaScript, SQL

Backend: Spring Boot, Spring Framework, Spring MVC, Spring Data JPA, Axum, Tokio, Tonic, NestJS, Node.js

Frontend: Next.js, Angular, React

APIs & Integration: REST, GraphQL, Apollo GraphQL, gRPC, Protocol Buffers, Third-Party API Integration

Databases & Caching: PostgreSQL, MySQL, DynamoDB, Redis, Amazon ElastiCache, Valkey

Architecture & Backend Engineering: Microservices, Distributed Systems, API Design, Data Modeling, GraphQL API Gateway, Asynchronous Workflows, Transactional Workflows

Security & Authentication: Authentication, Authorization, JWT, Refresh Tokens, RBAC, Session Management

Cloud, Messaging & DevOps: AWS, Amazon SQS, Amazon ECR, Amazon EKS, Docker, Jenkins, Argo CD, Grafana

Developer Tools: Git, Maven, Cargo, Postman, Altair GraphQL, RedisInsight, pgAdmin, NoSQL Workbench

PROFESSIONAL EXPERIENCE

Aretedge Innovations Pvt. Ltd.

Associate Software Developer

Hyderabad, Telangana, India (On-site)

Aug 2024 – May 2026

- Owned **90% of the Challenge and Casual Games modules end-to-end**, covering technical design, API/database design, business logic, integrations, deployment validation, and production support.
- Implemented game workflows for registration, participation, scoring, leaderboards, prize pools, Solana wallet integration, and cron-based Challenge lifecycle operations.
- Developed production backend functionality across **Java/Spring Boot and Rust microservices**, including APIs, service-layer logic, **asynchronous workflows**, database interactions, and service integrations.
- Developed and modified **20+ production APIs** using **REST, GraphQL, and gRPC** across Rust, Java, and NestJS backend services.
- Created and modified **Protocol Buffer contracts**, including service definitions, request/response messages, and enums, and implemented corresponding backend integrations using gRPC.
- Contributed to the **internal GraphQL API Gateway**, implementing schemas, models, resolvers/handlers, and service integrations connecting GraphQL operations with internal gRPC services.
- Worked within an **Axum, Tokio, and async-graphql Gateway architecture** supporting asynchronous HTTP/gRPC workflows across **distributed backend services**.
- Enhanced authentication workflows using OTP, JWT access tokens, refresh-token handling/rotation, RBAC, and Redis-backed session state.
- Created or significantly modified **6+ PostgreSQL tables and 12–15 functions/stored procedures**, using JSONB and materialized views for transactional and leaderboard/ELO workloads.
- Supported migration of approximately **20K user/login records** from DynamoDB to PostgreSQL using Node.js when batch/filter-heavy access patterns became inefficient for the workload.
- Used **Redis/ElastiCache** for authentication/session state and leaderboard workloads spanning daily, weekly, monthly, and country-based rankings across approximately **10–20 country variations**.
- Implemented end-to-end **Double Elimination tournament logic** covering winners/losers brackets, match progression, participant advancement, finals, and winner determination.

- Worked with **AWS SQS-based asynchronous application workflows** and AWS-hosted services including Aurora PostgreSQL, DynamoDB, ElastiCache/Valkey, ECR, and EKS-hosted applications.
- Independently handled approximately **4 Next.js applications** in a production white-labeled monorepo and developed Angular pages/components integrated with Java/Spring Boot APIs.
- Supported **25+ production releases, 10+ incidents, and 8+ hotfixes** through log analysis, root-cause investigation, API validation, Jenkins-triggered deployments, Docker build verification, and Argo CD-based application health checks.

Aspire Systems

Graduate Trainee / Software Engineering Intern

Chennai, Tamil Nadu, India (Remote)

Sep 2022 – Jul 2024

- Participated in software development projects through the Aspire Systems CoE program from September 2022 to July 2024, applying Java, Spring Boot, Angular, JavaScript, MySQL, REST APIs, Git, debugging, and application testing.
- Completed a formal Software Engineering Internship from January to March 2023, gaining hands-on experience in Java, MySQL, Angular, debugging, Git, and application testing.
- Built Angular applications using **reusable components, services, HttpClient, forms, routing, REST-style APIs, and CRUD workflows**, while applying Java/OOP concepts and MySQL database operations.

PROJECTS

Personal Finance Tracker | *Next.js 16, TypeScript, PostgreSQL, NeonDB, Zustand, Recharts*

Links: [Live Demo](#) | [GitHub](#)

- Built and deployed a full-stack finance application using **Next.js 16 App Router, TypeScript, and PostgreSQL**, React Server Components, and Server Actions.
- Implemented authentication/session management with **JWT, Argon2, HTTP-only cookies, and Edge Middleware**, with a multi-tenant PostgreSQL model using **RLS and explicit user scoping**.
- Built transaction workflows covering expenses, income, borrowing, repayment, lending, investments, adjustments, and investment-funded expenses.
- Developed analytics, reporting, and keyword-based automation for financial trends, categories, locations, contacts, investments, assets, lending, and debt using SQL and Recharts.
- Implemented CSV import/export with **versioned schemas, validation, category mapping, investment-balance checks, dry runs, and filtered exports**, and optimized PostgreSQL connection pooling for Vercel/Neon.

Student Management System | *Java, Angular 15, MySQL, JSON Server*

Links: [Live Demo](#) | [GitHub \(Angular\)](#) | [GitHub \(Java\)](#)

- Built a student-management application across **three technical iterations**, progressing from a JavaScript-based web application to an Angular 15 application and a Java CLI application backed by MySQL.
- Developed the Angular 15 web application using **reusable components, services, HttpClient, forms, and routing**, with JSON Server providing REST-style API endpoints for application data.
- Implemented the Java CLI version with MySQL, applying **object-oriented programming and relational database operations** for student-management workflows.

EDUCATION

K.S.R. College of Engineering

Bachelor of Engineering (B.E.) in Computer Science and Engineering | **CGPA: 8.56/10**

Namakkal, Tamil Nadu, India

Aug 2019 – May 2023

CERTIFICATIONS

Rust Programming Essentials – Edureka

AWS DynamoDB Fundamentals – Meta

Advanced React – Whizlabs

Google Prompting Essentials – Google